**Q1.** Every year thousands of trees are cut down in forests.

(a) Mammals and birds are two groups of animals that live in forests.

Give **two** reasons why fewer mammals and birds can survive after trees have been cut down.

1 .....

.....

1 mark

2 .....

.....

1 mark

(b) Many small plants grow in the clearings left after trees are cut down.

Explain why small plants are able to grow well after the trees have been cut down.

.....

.....

.....

2 marks

(c) In some forests, small branches are left on the ground.



Fungi and bacteria feed on these branches and release minerals, such as nitrates, back into the soil.

Why is it important that the minerals are released back into the soil?

.....

.....

1 mark

(d) A label was printed on the back of a birthday card.

> *The paper for this card was made from wood taken from sustainable forests.*

In sustainable forests, new trees are planted to replace trees that are cut down.

Give **two** reasons why it is important to replace forest trees that are cut down.

1 .....

.....

1 mark

2 .....

.....

1 mark

maximum 7 marks

## Q1.

(a) any **two** from

- less **or** no food
  *accept 'trees are producers'*
- fewer **or** no places to nest **or** shelter
  *accept 'they have lost their home'*
  *'they have lost their territory' is insufficient*
  *accept 'their habitat is destroyed'*
  *'it spoils the environment' is insufficient*
- less **or** no cover
  *accept 'they cannot hide from predators'*

**1 (L5)**

(b) any **one** from

- more light
- more water
- more space
- more nutrients
  *accept references to nutrients from trees left to rot*

**1 (L5)**

- more photosynthesis can take place
  *accept 'photosynthesis can take place'*

**1 (L6)**

(c) any **one** from

- to replace minerals **or** nitrates that have been removed (by plants)
  *accept 'minerals **or** nitrates get washed out of the soil'*
- plants absorb the minerals **or** nitrates
  *'plants feed on the nutrients' is insufficient*
- plants need minerals **or** nitrates to grow
  *accept 'so that plants can grow'*
  *accept 'plants can grow more'*
- minerals **or** nitrates **or** they are plant nutrients

**1 (L6)**

(d) any **two** from

- to provide food for animals
  *'it is good for the environment' is insufficient*

- to provide shelter **or** nesting sites **or** habitats for birds **or** animals

  *accept to provide homes for animals'*

  *'to help animals **or** birds survive' is insufficient*

- trees absorb carbon dioxide

- trees produce oxygen

- the roots bind soil **or** prevent soil washing away

  *accept provides trees to be cut down in the future'*
  *accept they need trees for paper' **or** 'to get paper'*

**2 (L5)**

**[7]**